Reproduction of the nodal rotation patch test and the convergence test of Rajendran (2010) for $\theta \geq 90°$ with the USEM code base

USEM-MATLAB study `studies/rajendran_panel`

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Abstract

The square-panel tests of Rajendran [1, Secs. 5.5–5.7] (mid-side nodes rotated by $\theta = 90°$–$270°$, overlapping and folded elements) are recomputed with the spectral elements of the code base on gmsh meshes. With the 9-node element defined as in the paper (parametric test functions, metric trial functions in Cartesian coordinates, 9th node at the centroid of the corners, 5×5 Gauss points), all values of the 9-node Tables 15–19 are reproduced to four digits, for the unsymmetric (PM-QUAD9*) as well as the conventional (PP-QUAD9) and the metric (MM-QUAD9) elements, except for entries that are inconsistent within the paper itself (Section 6). The energy-norm convergence rates of PM-QUAD9* (Fig. 15 of the paper) are reproduced as well (2.00 and 1.98 against 2.11 and 1.97). For $p \geq 3$ the unsymmetric spectral element reproduces the cubic field of the shear test on every mesh, including fully folded ones, up to round-off. The code base as is, however, places interior nodes with the 8-node map; on these meshes this makes interior and edge nodes coincide for even p, and the solution fails (Section 5).

1 Set-up

Panel $[0,2] \times [-1,1]$, plane stress, $\nu = 0.25$, three point supports ($u_x = 0$ at $(0, \pm 1)$, $u_x = u_y = 0$ at $(0,0)$), equilibrated loads on $x = 0$ and $x = 2$ (tension $\sigma_x = 1000$; moment $\sigma_x = -1500y$; shear $\sigma_x = -1500(2-x)y$, $\tau_{xy} = 750(1-y^2)$), displacement at $A = (2,0)$. The meshes (Fig. 1) are generated with gmsh (`mesh/gen_panel_meshes.py`): 2×2 super-elements of 2×2 elements; the mid-side nodes of the internal edges of every super-element are rotated about its centre by θ (Meshes a–f: $\theta = 0°, 90°, 135°, 175°, 225°, 270°$; sequences A, B, C: $\theta = 0°, 90°, 135°$ with 1×1, 2×2, 4×4 super-elements). For $\theta \geq 90°$ every element of every mesh has $\det \boldsymbol{J} \leq 0$ somewhere.

Material constant. The paper states $E = 1500$, but its exact solutions (1.0, 1.5, 3.3125) are those of $E = 2000$ (elasticity solutions with the supports above: 1.333, 2.0, 4.4167 for $E = 1500$; 1.0, 1.5, 3.3125 for $E = 2000$). $E = 2000$ is used here; all displacements scale with $1/E$.

Loads. The consistent nodal forces (`matlab/rp_loads.m`) agree with the lists of the paper (Fig. 8 and Sec. 5.6) except for the moment case of Fig. 8, where the paper lists -250, -500 at nodes 8, 9 (and 250, 500 at nodes 14, 13); the consistent, antisymmetric values are -500, -250 (and 500, 250). The listed values would give a moment of 1125 instead of 1000; since the paper's PM results equal the exact 1.5, the consistent values were evidently used.

Elements. With $p = 2$ (9 nodes): PP9 = conventional element; MM9 = metric functions on both sides, $\int \boldsymbol{B}_R^{\mathsf{T}} \boldsymbol{D} \boldsymbol{B}_R \det \boldsymbol{J}$; PM9 = PM-QUAD9*, $\int (\boldsymbol{B}_L^*)^{\mathsf{T}} \boldsymbol{D} \boldsymbol{B}_R$. As in the paper: metric functions Q_2 in Cartesian coordinates relative to the element centroid (`ShapeFunctions2D_Metric`, method 0), 9th node at the centroid of the corners, 5×5 Gauss points. In addition: "PM, skew" = the code-base metric functions (skew coordinates, method 1) with the same nodes; "PM, code" = `Ke_2D` with the nodes of `pSpectralElement_NodesTopology` (4×4 points). For $p = 3, \ldots, 8$: PP and PM with interior nodes from the bilinear corner map (the generalisation of the corner centroid) and $2p \times 2p$ points, and $\mathsf{PM}_{\mathrm{code}}$ with the interior nodes of the code base.

2 Nodal rotation patch test ($p = 2$)

PM9 passes the constant and the linear strain patch tests on all meshes, as in the paper (Tables 1 and 3); for the quadratic-strain (cubic displacement) shear test, which lies outside Q_2, its results are within 1.7 % of the exact value on all meshes (Table 2). All PP9 and MM9 values for moment and shear coincide with the paper

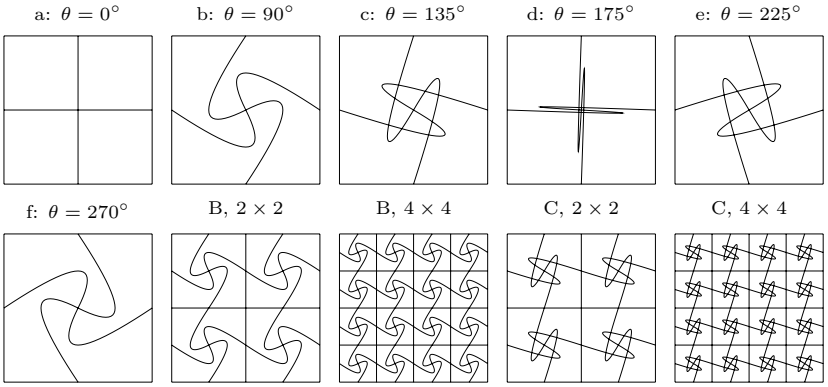


Figure 1: gmsh meshes: Meshes a–f of the patch test and the 2×2, 4×4 meshes of sequences B and C (cf. Figs. 9, 10 of the paper).

to four digits. The metric functions of the code base (skew coordinates) give the same results for the constant and linear strain tests and comparable ones for the shear test (3.15–3.41).

3 Convergence test (shear loading, $p = 2$)

The displacements of sequence A and of PP9 and MM9 for B and C agree with the paper. PM9 on sequence B yields the values the paper lists for sequence C and vice versa (Section 6). PM9 converges with the optimal rate 2 on the folded meshes of B and C, as reported in the paper. For $p = 3$ the unsymmetric element reproduces the field on all meshes (error $\leq 2 \cdot 10^{-12}$), whereas PP ($p = 3$) still has energy-norm errors of 9–80 %.

4 Higher degrees

With interior nodes from the bilinear corner map, the unsymmetric element is Jacobian-unlimited immune for $p = 3, \ldots, 8$ (Table 6): the cubic field is reproduced on all meshes with overlapping and folded elements. The round-off level grows with p and distortion (from 10^{-13} to 10^{-5} for $p = 8$, Mesh c), in line with the thick-cylinder study. The conventional element has errors of 7–100 % on the distorted meshes for all p.

5 A defect of the code base on these meshes

`pSpectralElement_NodesTopology` places the interior nodes with the 8-node (serendipity) map. On Meshes b and f this maps the interior node $\boldsymbol{\xi} = (0, 0)$ of the $p = 2$ element exactly onto a mid-side node of the same element (e.g. $(1, -0.5)$); the metric Vandermonde matrix is singular (cond $\approx 10^{18}$–10^{20}) and the computed displacements are zero ("code" columns of Tables 1 to 4). The same happens for every even p (interior nodes on $\xi = 0$ or $\eta = 0$), and on Meshes c–e the near-coincidence of nodes destroys the accuracy for $p \geq 4$ (Table 6). Placing the interior nodes with the bilinear map of the corners, which for $p = 2$ is the choice of the paper ("for simplicity"), removes the problem on all meshes tested. `ShapeFunctions2D_Metric` uses `decomposition`, which does not stop on a singular matrix, so the failure is silent.

6 Inconsistencies in the paper

1. $E = 1500$ is stated, the exact values correspond to $E = 2000$ (Section 1).
2. Fig. 8, moment loading: nodal forces at nodes 8/9 and 14/13 interchanged (Section 1).
3. Table 16, PP-QUAD9, Meshes b and f: 1.9088; the present value 3.0033 equals the paper's own PP-QUAD9 value for the same mesh in Table 18 (1×1 super-element of sequence B).
4. Tables 18 and 19, PM-QUAD9* columns: interchanged. The present values for sequence B (3.2710, 3.3000, 3.3110) are the paper's Table 19 entries (3.2710, 3.3001, 3.3110), those for C the Table 18 entries; the 1×1 values of Table 16 (b: 3.2710, c: 3.3044) confirm the present assignment.

Table 1: Moment loading, $u_y(A)$ (paper: Table 15; exact 1.5).

	PP9		MM9		PM9 (PM-QUAD9*)		PM, $p = 2$	
mesh	present	paper	present	paper	present	paper	skew	code
a (0°)	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000
b (90°)	1.2606	1.2606	1.7935	1.7935	1.5000	1.5000	1.5000	0.0000
c (135°)	1.0372	1.0372	1.6398	1.6398	1.5000	1.5000	1.5000	1.5000
d (175°)	1.2472	1.2472	1.5285	1.5285	1.5000	1.5000	1.5000	1.4996
e (225°)	1.0372	1.0372	1.6398	1.6398	1.5000	1.5000	1.5000	1.5000
f (270°)	1.2606	1.2606	1.7935	1.7935	1.5000	1.5000	1.5000	0.0000

Table 2: Shear loading, $u_y(A)$ (paper: Table 16; exact 3.3125). *: differs from the present result, see Section 6.

	PP9		MM9		PM9 (PM-QUAD9*)		PM, $p = 2$	
mesh	present	paper	present	paper	present	paper	skew	code
a (0°)	3.2979	3.2979	3.2979	3.2979	3.2979	3.2979	3.2979	3.2979
b (90°)	3.0033	1.9088*	3.9559	3.9559	3.2710	3.2710	3.3120	0.0000
c (135°)	-0.4993	-0.4993	3.8294	3.8294	3.3044	3.3044	3.4134	4.1224
d (175°)	2.5498	2.5498	3.3657	3.3657	3.3677	3.3677	3.1501	3.0340
e (225°)	-0.4993	-0.4993	3.8294	3.8294	3.3044	3.3044	3.4134	4.1224
f (270°)	3.0033	1.9088*	3.9559	3.9559	3.2710	3.2710	3.3120	0.0000

Table 3: Tension, $u_x(A)$ (paper: Table 14; exact 1.0).

	PP9		MM9		PM9 (PM-QUAD9*)		PM, $p = 2$	
mesh	present	paper	present	paper	present	paper	skew	code
a (0°)	1.0000	1.0	1.0000	1.0	1.0000	1.0	1.0000	1.0000
b (90°)	1.0000	1.0	1.1300	1.0*	1.0000	1.0	1.0000	0.0000
c (135°)	1.0000	1.0	1.4633	1.0*	1.0000	1.0	1.0000	1.0000
d (175°)	1.0000	1.0	1.0285	1.0*	1.0000	1.0	1.0000	0.9995
e (225°)	1.0000	1.0	1.4633	1.0*	1.0000	1.0	1.0000	1.0000
f (270°)	1.0000	1.0	1.1300	1.0*	1.0000	1.0	1.0000	0.0000

5. Table 14, MM-QUAD9: 1.0 for all meshes. With a stiffness matrix that reproduces all MM9 moment and shear values to four digits, MM9 gives 1.1300 (b, f), 1.4633 (c, e), 1.0285 (d), also with load vectors computed from the metric functions (Eq. (22) of the paper). MM9 does not pass the constant strain patch test on these meshes; the paper's entries could not be reproduced.

Apart from these items, every 9-node value of the paper for $\theta \geq 90°$ is reproduced to four digits. The 8-node results (Tables 8–13) are not reproduced: the code base provides only tensor-product (Lagrange) elements.

7 Reproduction

`mesh/gen_panel_meshes.py` (gmsh), `matlab/rp_run.m` (all computations, GNU Octave 8.4 or MATLAB, a few minutes), `postprocess.py` (tables). Paper values: `results/paper_values.csv`, `results/paper_rates.csv`.

References

[1] S. Rajendran. A technique to develop mesh-distortion immune finite elements. *Computer Methods in Applied Mechanics and Engineering*, 199:1044–1063, 2010. doi: 10.1016/j.cma.2009.11.017.

Table 4: $u_y(A)$ for the sequences A, B, C (paper: Tables 17–19; exact 3.3125).

		PP9		MM9		PM9		PM, $p=2$	
seq.	mesh	present	paper	present	paper	present	paper	skew	code
A	1×1	3.2979	3.2979	3.2979	3.2979	3.2979	3.2979	3.2979	3.2979
	2×2	3.3106	3.3106	3.3106	3.3106	3.3106	3.3106	3.3106	3.3106
	4×4	3.3123	3.3123	3.3123	3.3123	3.3123	3.3123	3.3123	3.3123
B	1×1	3.0033	3.0033	3.9559	3.9559	3.2710	3.3044*	3.3120	0.0000
	2×2	3.1768	3.1768	3.9519	3.9519	3.3000	3.3057*	3.3185	0.0000
	4×4	3.2662	3.2661	3.9598	3.9598	3.3110	3.3116*	3.3121	0.0000
C	1×1	-0.4993	-0.4993	3.8294	3.8294	3.3044	3.2710*	3.4134	4.1224
	2×2	2.9218	2.9218	3.9208	3.9208	3.3057	3.3001*	3.2600	3.2935
	4×4	3.1531	3.1531	3.9660	3.9660	3.3116	3.3110*	3.3067	3.3105

Table 5: Least-squares rates of the energy-norm strain error, Eq. (24) of the paper (paper: Fig. 15). PP on folded meshes: the parametric strain is singular where $\det \boldsymbol{J} = 0$, so the energy error of PP is not well defined for B and C (here: signed $\det \boldsymbol{J}$). "–": error at round-off level (field reproduced).

	PP9		PM9		PM, $p=2$	PM, $p=2$	PP, $p=3$	PM, $p=3$
seq.	present	paper	present	paper	skew	code		
A	1.99	1.93	1.99	1.93	1.99	1.99	–	–
B	0.70	1.05	2.00	2.11	2.02	0.01	0.64	–
C	2.50	0.85	1.98	1.97	2.13	2.41	0.94	–

Table 6: Shear loading on Meshes a–f: relative nodal error e_n for $p = 3,\dots,8$ (cubic field, contained in the PM trial space for $p \geq 3$). PM: interior nodes from the bilinear corner map; $\mathsf{PM}_{\text{code}}$: interior nodes of `pSpectralElement_NodesTopology`.

	a (0°)	b (90°)	c (135°)	d (175°)	e (225°)	f (270°)
PP, $p=3$	5e-14	2e-1	1e-1	2e-1	1e-1	2e-1
PM, $p=3$	2e-14	8e-14	1e-13	3e-13	5e-14	2e-13
$\mathsf{PM}_{\text{code}}$, $p=3$	6e-14	2e-13	7e-10	1e0	2e-9	1e-12
PP, $p=4$	2e-13	8e-2	5e-1	2e-1	5e-1	8e-2
PM, $p=4$	1e-13	5e-13	8e-12	5e-10	2e-11	1e-13
$\mathsf{PM}_{\text{code}}$, $p=4$	8e-14	1e0	4e-7	2e1	5e-7	1e0
PP, $p=5$	8e-13	7e-2	9e-2	2e-1	9e-2	7e-2
PM, $p=5$	6e-13	1e-10	2e-12	2e-10	1e-12	7e-11
$\mathsf{PM}_{\text{code}}$, $p=5$	6e-13	1e-9	1e-5	1e0	1e-4	4e-9
PP, $p=6$	5e-13	9e-2	2e-1	1e-1	2e-1	9e-2
PM, $p=6$	5e-14	2e-11	5e-11	2e-10	8e-11	7e-12
$\mathsf{PM}_{\text{code}}$, $p=6$	4e-13	1e0	6e-3	1e0	8e-2	1e0
PP, $p=7$	2e-13	1e-1	9e-2	1e-1	9e-2	1e-1
PM, $p=7$	3e-14	3e-11	1e-9	4e-9	4e-10	5e-11
$\mathsf{PM}_{\text{code}}$, $p=7$	6e-13	1e-6	1e1	1e0	4e0	1e-6
PP, $p=8$	3e-12	1e-1	1e0	1e-1	1e0	1e-1
PM, $p=8$	7e-13	3e-9	1e-5	2e-7	3e-6	3e-9
$\mathsf{PM}_{\text{code}}$, $p=8$	1e-12	1e0	8e2	1e0	1e3	1e0